

MPG No. 8.4.27 $m'm'm$ (m'm'm setting) [Type III, orthorhombic] [G tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$\begin{aligned} G_{yzx} &= 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\ G_{zxy} &= -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \end{aligned}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$\begin{aligned} G_{yzx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)} \\ G_{zxy} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)} \\ G_{xyz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)} \end{aligned}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).